

TITLE PROPOSAL 1

PROPOSED TITLE: Rent & Play

I. PROJECT OVERVIEW AND RATIONALE

Rent & Play is a small rental business in Lakewood Subd., Los Baños, Laguna, managed by Reynold Pastor. The business rents sports equipment, board games, card games, and gaming consoles. The current rental process involves item availability checking, customer information recording, rental recording, due-date monitoring, return condition checking, and fee or penalty calculation.

The verified problem is the time and effort required to manage these activities using paper-based records and manual checking. It can be difficult to quickly confirm whether an item is available, rented, or under maintenance, and it also creates extra work when checking due dates and charges. The revised proposal adds an ESP32-assisted verification terminal so that a transaction is not only saved in the system but also physically displayed and confirmed at the rental counter.

Rent & Play proposes a web dashboard and mobile application sharing one backend/API and database. The mobile application will scan the equipment QR code and submit the rental or return request. After the backend validates the request, the ESP32 terminal receives the transaction status through Wi-Fi API integration. It then displays the status on an OLED screen, uses an LED and buzzer for notification, and waits for a physical button press to confirm the transaction. This creates one connected workflow across Mobile, Backend/API, Database, Web, and ESP32. The ESP32 is not a passive display: no rental or return transaction is marked Confirmed in the database until the physical button on the terminal is pressed. This makes the ESP32 a functional confirmation gate over the release/return process.

II. STATEMENT OF THE PROBLEM

How can Rent & Play improve the rental and return process by reducing manual recording, making equipment status easier to verify, and providing a clear physical confirmation of each transaction through the mobile application and ESP32 terminal?

1. Manual rental and return recording – Rental and return details are currently recorded and checked manually, which can cause delays, missing information, or incorrect transaction records.
2. Unclear equipment status – It can be difficult to quickly determine whether an item is available, rented, or under maintenance because the status depends on manual checking and updating.
3. Lack of immediate transaction confirmation – After scanning an equipment QR code and submitting a rental or return transaction, there is no dedicated physical device that clearly confirms whether the transaction is successfully accepted, rejected, or still pending.
4. Limited handling of verification errors – Problems such as failed Wi-Fi/API connection, duplicate confirmation, or an invalid transaction are not clearly handled in the current manual process, which may lead to confusion during equipment release and return.



III. PROJECT OBJECTIVES

General objective

To develop an integrated web, mobile, backend/database, and ESP32-assisted rental and return management system for Rent & Play that centralizes equipment inventory, customer information, rental and return transactions, due dates, configured rates, deposits, applicable penalties, item conditions, and maintenance status, while using mobile QR-code scanning and an ESP32 verification terminal to provide a controlled, organized, and physically confirmed transaction process.

Specific objectives

1. Digitally manage rental items, customer information, transactions, due dates, configured rental rates, deposits, penalties, conditions, and maintenance status.
2. Use the mobile application camera to scan equipment QR codes and retrieve the correct item record before rental or return processing.
3. Validate item availability and prevent conflicting rental requests before creating a Pending Verification transaction; finalize the item status only after the ESP32 terminal confirms the rental or return through the physical button.
4. Allow the Owner/Operator to configure rental rates and deposits, and calculate applicable rental fees and late penalties using the configured business rates and rules.
5. Use the ESP32 terminal to display transaction status on an OLED, provide LED/buzzer feedback, and send button-based confirmation to the backend through Wi-Fi.
6. Maintain searchable rental history and return-condition records while recording the verification result, terminal ID, and confirmation time for each completed transaction.

IV. CLIENT VALUE AND SUCCESS INDICATORS

Expected client outcome	Current baseline/evidence	Target or acceptance criteria indicator
Faster item lookup	Operator manually checks paper records.	Mobile app opens the correct item record after scanning its QR code.
Better due-date visibility	Due dates are manually monitored.	Active rentals show due dates and approaching/overdue status in the dashboard.
More consistent return records	Return condition and charges are manually recorded.	Release/return, condition, deposit, penalty, and verification status are stored per transaction.
Physical transaction verification	There is no connected device that confirms a transaction at the counter.	ESP32 receives the backend status, displays it on OLED, alerts through LED/buzzer, and records button confirmation.



TITLE PROPOSAL 1 - SCOPE AND USERS

V. SCOPE, LIMITATIONS, AND EXCLUSIONS

In scope	Out of scope / limitation
Rental inventory; customer records; mobile QR-code scanning; rental and return transactions; due-date tracking; overdue alerts; configured fees/deposits/penalties; return and condition logs; maintenance status; rental history.	No guaranteed ID authenticity; no automated damage detection; no continuous item tracking outside business records; no supplier ordering or external marketplace; no online payment gateway.
Web management dashboard supporting the mobile operator workflow.	Mobile QR-code scanning depends on readable equipment labels and a working device camera.
REST API, shared database, and ESP32 Wi-Fi terminal.	ESP32 is not an autonomous scanner and does not replace the mobile QR-scanning function.
ESP32 terminal with OLED, LED, buzzer, physical confirmation button, and Wi-Fi communication.	Prototype is intended for semester demonstration; no production deployment or unnecessary sensitive-data collection.
Sample or approved business data for testing.	No guarantee of operation during internet outage; the terminal should fail safely and avoid confirming transactions without a valid backend response.

VI. PROJECT LOCALE AND STAKEHOLDERS

Stakeholder / user	Role in current process	Expected use of proposed system
Business Owner/Operator	Manages inventory, customer records, rates, rentals, returns, maintenance, and reports.	Uses the mobile app to scan equipment QR codes, submit rental/return requests, review item status, and complete physical terminal confirmation; uses the web dashboard to manage and monitor inventory, rates, rentals, fees, due dates, maintenance, transaction history, and ESP32 verification results.
Customer / Renter	Provides required information to the Owner/Operator and receives rental/return details.	Provides required information, reviews rental terms, and receives the confirmed rental/return transaction details. The customer does not directly operate the system.
ESP32-Assisted Terminal	Physical verification device located at the rental counter.	Connects to Wi-Fi API, displays the transaction status on OLED, gives LED/buzzer feedback, and sends button confirmation back to the backend.



TITLE PROPOSAL 1 - INTEGRATION DESIGN

VII. PROPOSED INTEGRATED WORKFLOW

1. Equipment Identification & Transaction Initiation The workflow begins when the Owner/Operator selects either Rental or Return in the mobile application and scans the equipment QR code using the phone camera. The QR code provides the unique equipment ID. The mobile app uses this ID to retrieve the item's current status and information from the Backend/API. The Owner/Operator then provides or confirms the required customer and transaction details before submitting the request.

2. Transaction Validation & Pending Verification Once submitted, the transaction data is transmitted to the Backend/API, where the system checks item availability, customer information, transaction type, and applicable rental or return rules. The backend also checks for conflicting active or pending transactions for the same item. If the request is valid, the system creates a transaction with a Pending Verification status and temporarily reserves the item from conflicting rental requests until the transaction is completed, rejected, or expired. The transaction details are then sent through Wi-Fi to the assigned ESP32 verification terminal.

3. ESP32 Transaction Verification The ESP32 terminal receives the pending transaction and displays the relevant information on the OLED screen, such as the transaction type, equipment ID, and ready status. The Owner/Operator checks the displayed information against the actual equipment and transaction details. For a return, the Owner/Operator also checks the physical condition of the equipment. After confirming that the information is correct, the Owner/Operator presses the confirmation button on the ESP32 to authorize the transaction.

4. API Confirmation & Transaction Result After the confirmation button is pressed, the ESP32 sends the transaction ID back to the Backend/API through Wi-Fi. The backend verifies that the transaction is still valid and has not expired, been duplicated, or already completed. When verification is successful, the system marks the transaction as Confirmed; otherwise, it rejects the transaction and returns the appropriate error status to the ESP32 and mobile application. A rejected or expired rental releases the temporary reservation on the item, while a rejected or expired return leaves the item Rented.

5. Equipment Status & Record Synchronization For a confirmed rental, the system updates the equipment status to Rented and records the due date, rental fee, deposit, and other transaction details. The rental fee and deposit are based on the configured rates maintained by the Owner/Operator. For a confirmed return, the system records the return time provided in the request and equipment condition, calculates any applicable late penalty, and changes the item status to Available or Under Maintenance when necessary. These changes are stored in the shared database and synchronized with the mobile application and web dashboard.

6. Physical & Digital Confirmation The ESP32 provides an immediate physical confirmation by showing the final status on the OLED and activating the corresponding LED and buzzer. A successful transaction displays a confirmation



message with a success indication, while a failed or rejected transaction displays an error message. At the same time, the mobile application and web dashboard reflect the verified transaction. This gives the Owner/Operator a consistent record of equipment status and transaction history.

VIII. PLATFORM RESPONSIBILITY MATRIX

Component	Primary responsibility	Key inputs	Key outputs
Web application	Management dashboard for inventory, customers, active rentals, due dates, fees, maintenance, reports, and device/verification history.	Rental/inventory data, customer records, conditions, configured rates/deposits, transaction records, and device status.	Dashboard, alerts, reports, management actions, verification history.
Mobile application	On-site rental and return workflow; scans equipment QR codes and submits transaction requests without setting final item status.	QR scan result, customer details, rental/return data.	Pending transaction request, transaction result, and updated item status.
Backend / REST API	Authentication, validation, business rules, rate/fee calculation, transaction state transitions, API integration, and device synchronization.	Mobile/web requests and ESP32 confirmation.	Validated transaction records, device instructions, final status responses.
Shared database	Central source of truth for items, customers, transactions, configured rates/deposits, due dates, conditions, maintenance, terminal records, and history.	Validated backend data.	Current item/rental state and audit/verification history.
ESP32 terminal	Physical rental/return verification endpoint connected through Wi-Fi.	Pending transaction status from API; button input.	OLED status, LED/buzzer feedback, button confirmation sent to API.
External service / network	Provides Wi-Fi connectivity and hosting for the backend/database where applicable.	HTTPS API traffic.	Network access and persistent service communication.



TITLE PROPOSAL 1 - DATA, DEVICE, AND FEASIBILITY

IX. PRELIMINARY API AND DATA CONTRACT

Flow / endpoint	Method or protocol	Example payload / shared record	Consumer
POST /api/rentals	HTTPS JSON	{ "customerId": 10, "itemId": "PS4-01", "dueDate": "2026-09-04T18:00:00", "terminalId": "TERM-01" } Backend applies the configured rental rate and deposit and creates the Pending Verification record.	Mobile -> Backend
POST /api/returns	HTTPS JSON	{ "rentalId": 45, "returnedAt": "2026-09-04T20:00:00", "condition": "Good", "terminalId": "TERM-01" } Backend calculates any applicable penalty and creates the Pending Verification record.	Mobile -> Backend
GET /api/terminals/{terminalId}/pending	HTTPS JSON	{ "transactionId": 45, "type": "RETURN", "itemId": "PS4-01", "displayStatus": "RETURN READY" }	ESP32 -> Backend
POST /api/terminals/{terminalId}/confirm	HTTPS JSON	{ "transactionId": 45, "action": "CONFIRM", "button": "OK", "confirmedAt": "2026-09-04T20:01:10" }	ESP32 -> Backend
PATCH /api/items/{id}/status	HTTPS JSON	Internal backend update: { "status": "Rented" }, { "status": "Available" }, or { "status": "Under Maintenance" }	Backend
GET /api/terminals/{terminalId}/status	HTTPS JSON	{ "terminalId": "TERM-01", "online": true, "lastSeen": "2026-09-04T20:01:10" }	Web -> Backend

X. ESP32 ROLE AND HARDWARE PLAN

Decision: ☒ Required because it provides physical transaction verification and user feedback ☐ Optional enhancement ☐ Not applicable

Sensor/input	Actuator/output	Sampling or trigger rule	Offline/error behavior
ESP32 + Wi-Fi connection	OLED display shows RENT READY, RETURN READY, CONFIRMED, REJECTED, or OFFLINE.	Connect to Wi-Fi and poll the terminal API for pending transactions at a short interval. Refresh when a new status is received.	If Wi-Fi/API is unavailable, show OFFLINE/API ERROR and do not confirm the transaction locally.
Push button	Confirmation request is sent to the backend API.	Accept input only when a valid pending rental or return is displayed; one valid press per transaction.	Ignore presses with no pending transaction and reject duplicate confirmations.
Backend transaction status/result	LED and buzzer provide visual and audible feedback for ready, success, and error states.	Trigger when a pending transaction arrives and when the API returns confirmation or rejection.	Use the error state for rejected/expired requests; avoid continuous alarms.
Registered terminal ID + device status	Backend receives terminal last-seen/status information for monitoring.	Send status during normal communication and after reconnection.	If disconnected, keep the transaction Pending Verification until the terminal reconnects; do not confirm locally.

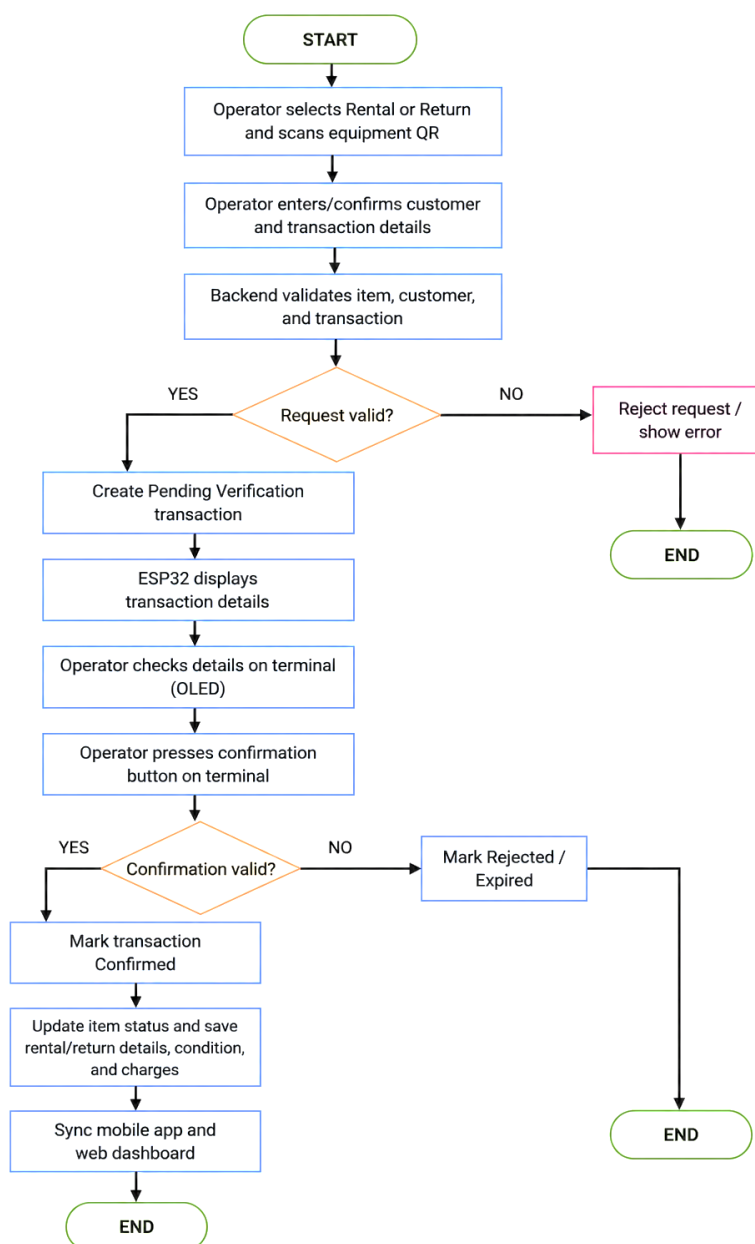


XI. CLIENT CONFIRMATION FOR THIS ALTERNATIVE

The client representative confirms that the problem and proposed scope above were discussed. This is preliminary validation and does not authorize production deployment or financial commitment.

Client representative / signature	Date	Comments or required revisions
 Reynold Pastor	August 21, 2026	

XII. FLOWCHART





TITLE COMPARISON AND RECOMMENDATION

Score each title using the same evidence. A feasible, integrated, client-valuable project should rank higher than a larger but weakly connected feature set.

Criterion	Weight	Title 1	Title 2	Title 3
Client need and evidence	20%			
Integration depth and coherence	25%			
Web/Mobile role clarity	15%			
ESP32 relevance and feasibility	15%			
Semester feasibility and resources	10%			
Security, privacy, and ethical fit	10%			
Client willingness and access	5%			
TOTAL	100%			

TEAM RECOMMENDATION

Recommended title: Rent & Play- QR-Based Rental, Due Date, and Fee Management System

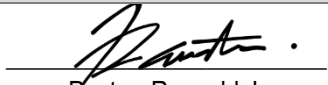
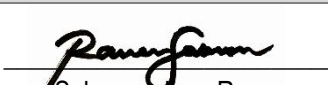
Justification based on the comparison, client feedback, feasibility, and integration value:

PROPOSED TEAM RESPONSIBILITIES

Member	Primary ownership	Secondary/cross-training responsibility	Evidence of contribution
Pastor, Reynold J.			
Saberon, Jhon Raven			

DECLARATION

We certify that the information in this proposal is based on our preliminary investigation and client consultation. We understand that title approval does not authorize collection of sensitive data, production deployment, expenses, or use of client resources beyond written permissions.

Student name and signature	Student name and signature
 Pastor, Reynold J.	 Saberon, Jhon Raven